

CALAPAN, Philippines

WMO: 984310

Lat: 13.417N

Lon: 121.1868E

Elev: 135

StdP: 14.62

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.0	73.8	67.4	101.1	75.6	68.6	105.7	76.1	13.7	77.8	12.1	78.0	3.2	90	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.5	91.4	82.4	90.3	81.9	89.4	81.6	83.8	89.4	83.0	88.7	82.4	88.2	4.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.3	169.1	88.2	81.4	164.2	87.6	80.7	160.4	87.1	48.1	89.7	47.2	89.1	46.4	88.4	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.2	10.5	9.0	DB	69.0	94.6	4.1	3.1	66.0	96.9	63.6	98.7	61.3	100.5	58.3	102.8
			WB	66.7	85.3	3.0	1.1	64.5	86.0	62.8	86.6	61.1	87.2	58.9	88.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.4	78.5	79.3	80.8	83.1	83.9	83.0	82.0	82.4	82.0	81.6	80.9	79.4
	DBStd	2.43	1.71	1.73	1.71	1.82	1.79	1.86	2.01	2.11	1.83	1.74	1.72	1.93
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11467	885	821	956	993	1050	990	991	1003	960	978	928	912
	CDD65	5992	420	401	491	543	585	540	526	538	510	513	478	447
	CDH74	60728	3067	3232	4701	6211	6997	6176	5545	5828	5353	5261	4653	3704
	CDH80	17454	351	534	1118	2185	2773	2243	1709	1881	1621	1467	1048	524
(14) Wind	WSAvg	3.3	3.9	3.9	4.1	3.8	3.0	2.6	2.6	2.7	2.8	2.9	3.3	4.1
(15) Precipitation	PrecAvg	74.2	3.4	1.9	2.0	3.1	6.9	6.8	8.2	7.3	7.1	10.4	9.4	7.2
	PrecMax	100.3	8.5	7.6	6.3	11.3	18.4	13.3	12.3	12.8	19.2	21.5	38.4	16.1
	PrecMin	40.8	0.9	0.4	0.0	0.5	1.2	0.9	0.9	1.0	1.5	4.2	1.7	2.1
	PrecStd	16.2	2.1	1.9	1.6	2.8	4.9	3.4	3.3	3.5	4.9	5.2	8.4	4.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.7	87.1	88.7	91.4	93.3	92.1	91.0	91.7	91.0	90.0	88.5	86.4
		MCWB	78.6	79.2	80.0	81.7	83.6	83.0	82.7	82.8	82.7	81.9	81.6	80.5
	2%	DB	84.2	85.4	87.2	89.9	91.4	90.7	89.4	89.9	89.1	88.4	86.9	85.2
		MCWB	78.2	78.3	79.3	80.9	82.3	82.4	81.9	82.0	81.8	81.2	80.8	79.8
	5%	DB	83.2	84.4	86.2	88.8	90.1	89.5	88.2	88.5	88.0	87.2	85.9	84.1
		MCWB	77.6	77.8	78.8	80.4	81.7	82.0	81.3	81.4	81.3	80.7	80.3	79.1
	10%	DB	82.1	83.3	85.1	87.7	89.0	88.2	86.9	87.2	86.8	86.0	84.8	82.9
		MCWB	77.1	77.3	78.3	79.9	81.3	81.4	80.8	80.8	80.9	80.3	79.7	78.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.0	80.6	81.7	83.5	84.6	84.6	84.0	84.4	84.3	83.3	83.3	81.8
		MCDB	84.1	85.4	86.9	89.4	91.3	90.1	89.2	89.7	89.4	88.0	87.2	85.4
	2%	WB	79.0	79.4	80.5	82.3	83.4	83.3	82.6	82.8	82.8	81.9	81.6	80.4
		MCDB	83.0	84.2	86.0	88.6	90.0	89.2	88.0	88.5	88.0	87.0	85.8	84.3
	5%	WB	78.2	78.6	79.6	81.3	82.5	82.4	81.8	81.9	81.8	81.2	80.7	79.6
		MCDB	82.1	83.3	84.9	87.4	88.9	88.4	87.1	87.5	87.0	86.1	85.0	83.1
	10%	WB	77.5	77.8	78.8	80.5	81.7	81.7	81.0	81.0	81.0	80.5	80.0	78.8
		MCDB	81.1	82.2	83.9	86.4	87.8	87.4	86.2	86.3	86.0	85.3	84.0	82.1
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.9	8.7	9.5	10.5	10.3	9.5	9.6	9.7	9.3	7.9	6.7
		MCDBR	8.4	9.3	9.9	10.9	12.3	12.0	11.3	11.6	11.6	11.0	9.2	8.3
		MCWBR	5.1	5.4	5.3	5.4	6.5	6.9	6.5	6.6	6.7	6.3	5.5	5.0
	5% WB	MCDBR	7.8	8.9	9.5	10.5	11.8	11.5	10.9	11.1	11.0	10.4	9.0	7.9
		MCWBR	5.2	5.7	5.8	6.1	6.9	7.1	6.8	6.9	7.0	6.5	5.7	5.3
(40) Clear-Sky Solar Irradiance	taub	0.431	0.423	0.444	0.455	0.452	0.445	0.442	0.448	0.446	0.468	0.443	0.450	(40)
	taud	2.410	2.420	2.361	2.336	2.372	2.411	2.417	2.392	2.391	2.312	2.392	2.353	(41)
	Ebn at Noon	275	283	278	272	269	268	269	271	273	266	269	266	(42)
	Edh at Noon	37	37	41	42	39	37	37	39	39	41	37	38	(43)
(44) All-Sky Solar Radiation	RadAvg	1385	1649	1858	2052	1846	1662	1503	1484	1526	1478	1401	1244	(44)
	RadStd	110	157	142	111	168	138	171	118	157	102	122	112	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.45	+0.70	N/A	+0.79	+0.65	+0.81	N/A	N/A	+284	+284

Nomenclature: See separate page